



C23-CM-403

23514

BOARD DIPLOMA EXAMINATION, (C-23)

MARCH/APRIL—2025

DCME – FOURTH SEMESTER EXAMINATION

COMPUTER ORGANIZATION AND MICROPROCESSORS

Time : 3 hours]

[Total Marks : 80

PART—A

3×10=30

- Instructions :**
- (1) Answer **all** questions.
 - (2) Each question carries **three** marks.
 - (3) Answers should be brief and straight to the point and shall not exceed five simple sentences.

- 1. Write the purpose of (i) Program counter and (ii) Memory address register.
- 2. Draw the functional block diagram of a simple accumulator based CPU.
- 3. List any six addressing modes.
- 4. What is the fixed point representation of a real number? Write any two reasons to choose fixed point representation.
- 5. Why do we need memory hierarchy in a computer?
- 6. Write any three differences between main memory and auxiliary memory.
- 7. Define an interface.
- 8. List different asynchronous data transfer modes.
- 9. What is a microprocessor? List any two companies that develop microprocessors.
- 10. Write the differences between RISC and CISC in terms of number of operations within an instruction, instruction size and number of clock cycles for the execution of an instruction.

PART—B

10×5=50

- Instructions :**
- (1) Answer *any five* questions.
 - (2) Each question carries **ten** marks.
 - (3) Answers should be comprehensive and the criteria for valuation is the content but not the length of the answer.

- 11. Explain Flynn's classification of computers.
- 12. Convert $(5.75)_{10}$ to binary and normalize to store in 32 bit IEEE 754 standard.
- 13. Explain instruction formats with an example for each instruction format.
- 14. Explain virtual address and physical address organization.
- 15. Briefly explain the memory device characteristics : (i) Access time, (ii) Access rate, (ii) Alterability, (iv) Permanence of storage and (v) Cycle time.
- 16. Explain interrupt initiated data transfer method.
- 17. Explain strobe control and hand shaking method of data transfer.
- 18. Explain functional block diagram of 8086 microprocessor.

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